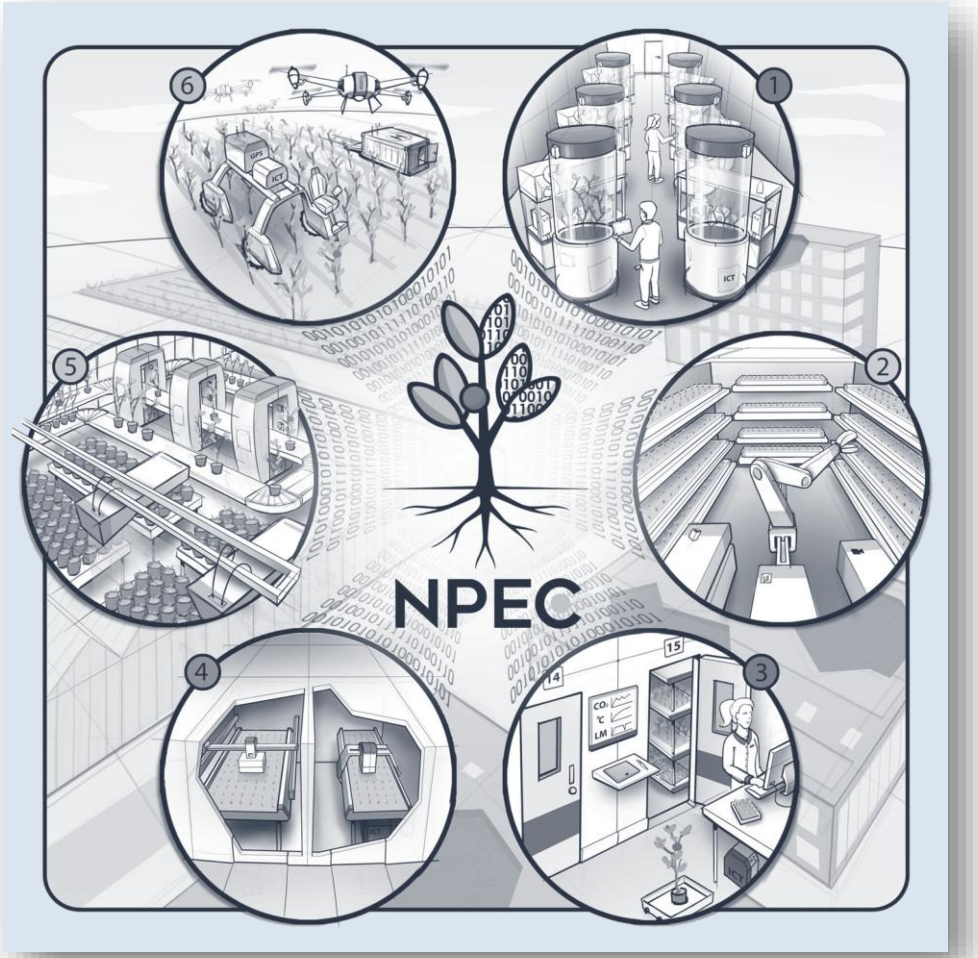
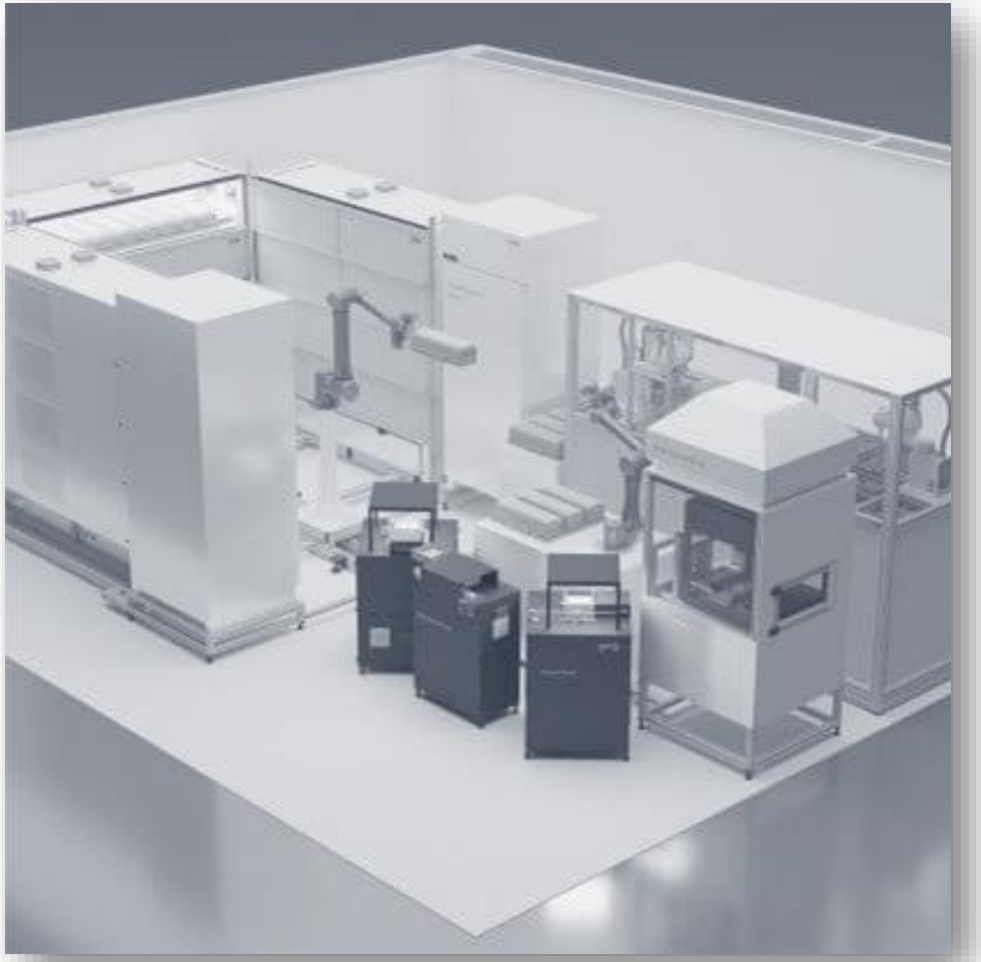
End-to-End Inoculation

Stijn Korver (242824)

Netherlands Plant Eco-phenotyping Centre





Plant-Microbe Interaction

Hades

Proposed Solution

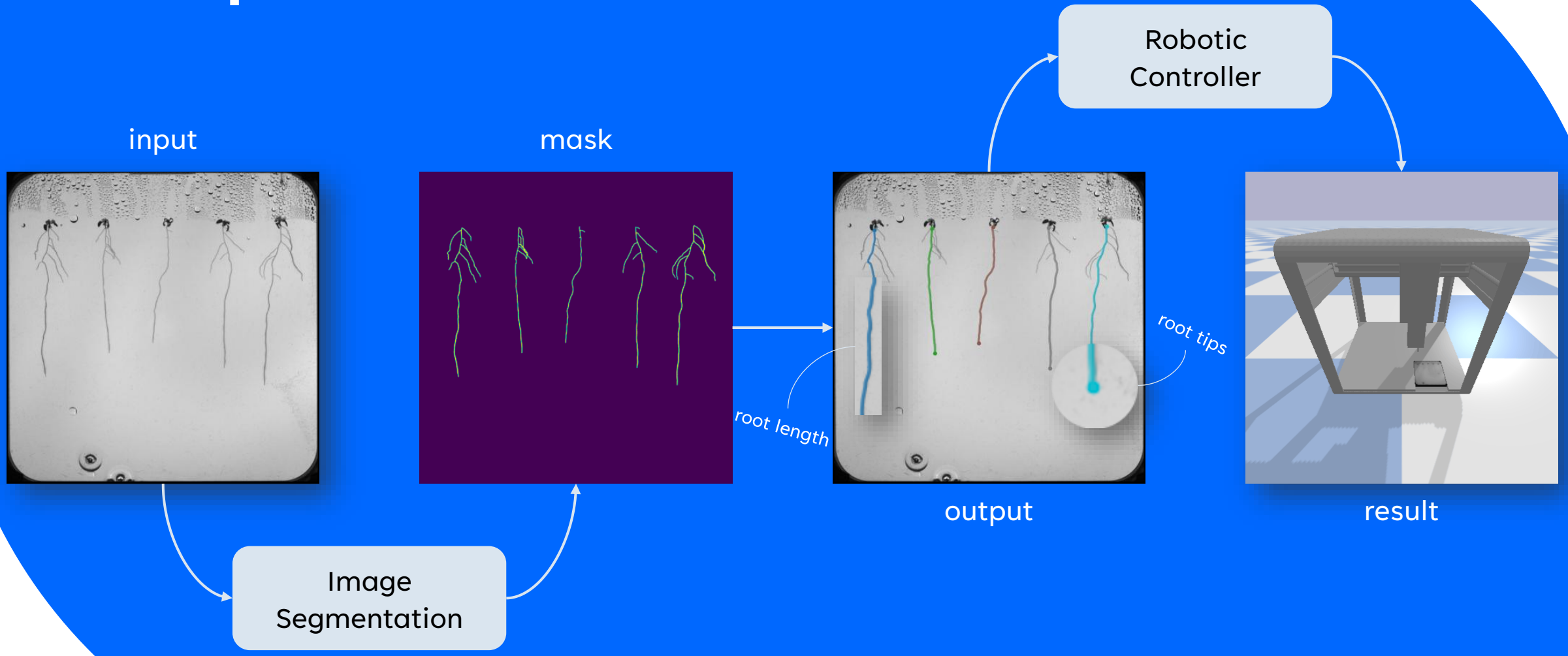
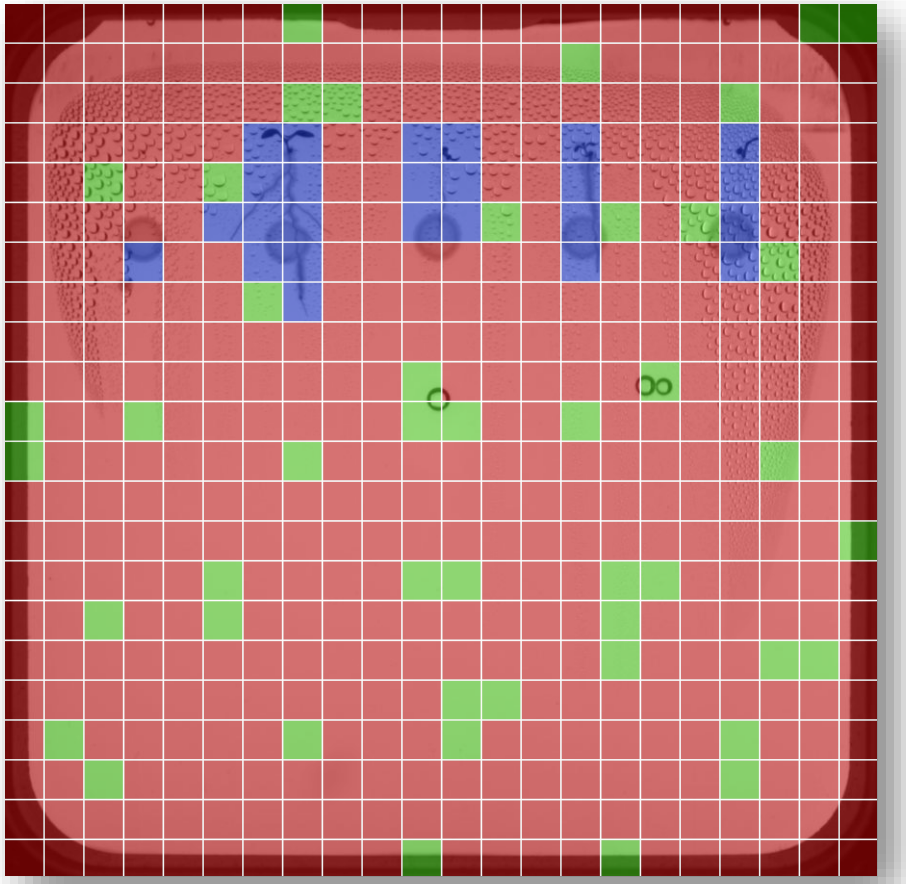


Image Segmentation Model



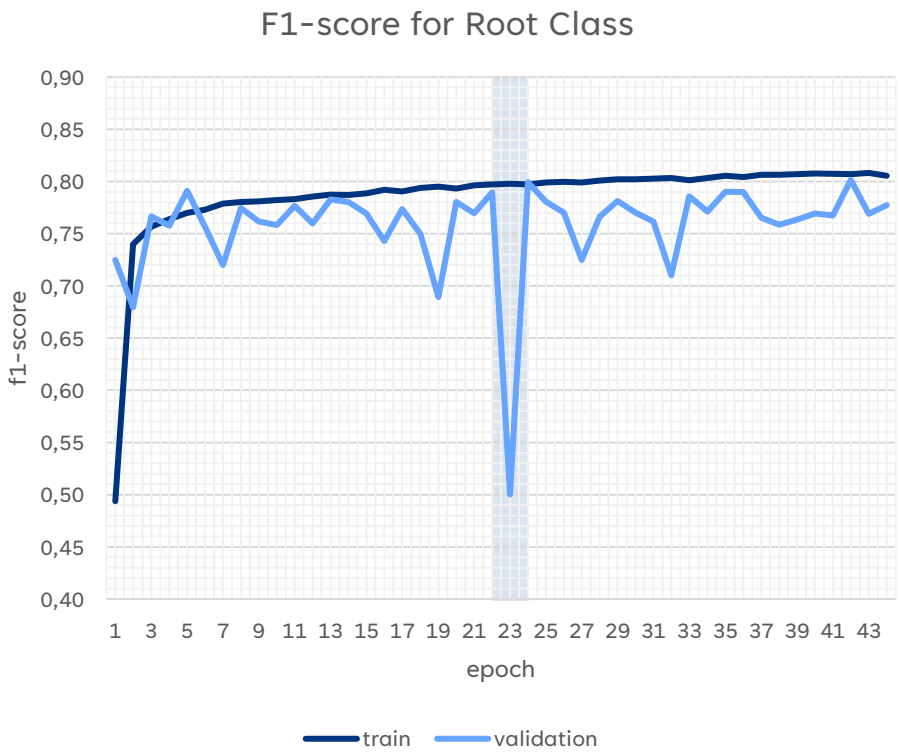
128x128
patch size

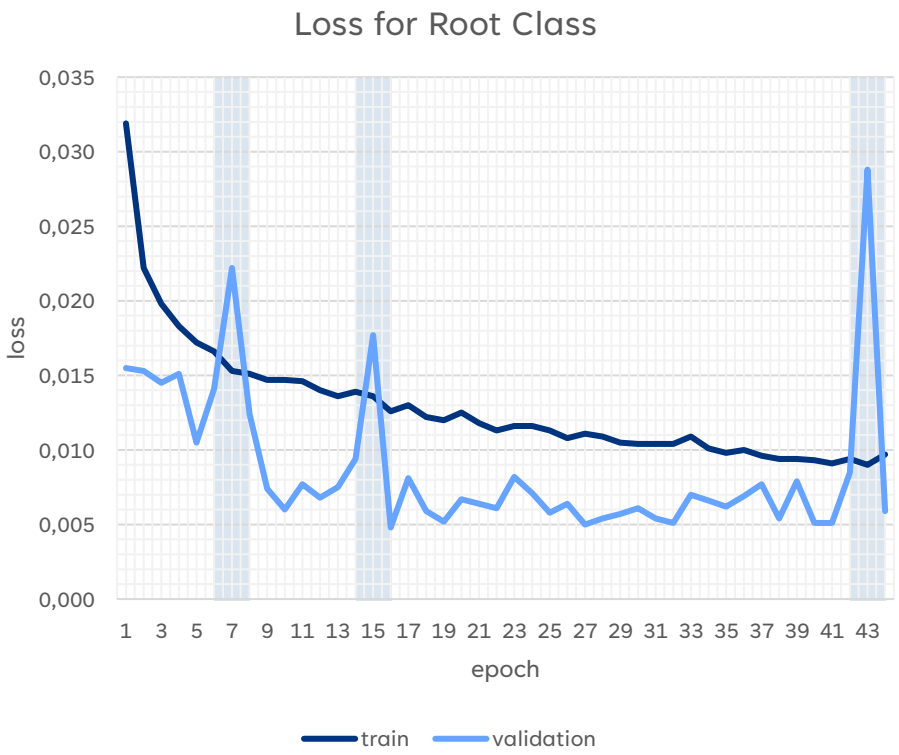
Background-Only
90% removed

Background-Only
10% kept

Root, Seed, Shoot
100% kept

Image Segmentation Model





Segmentation Model Iterations

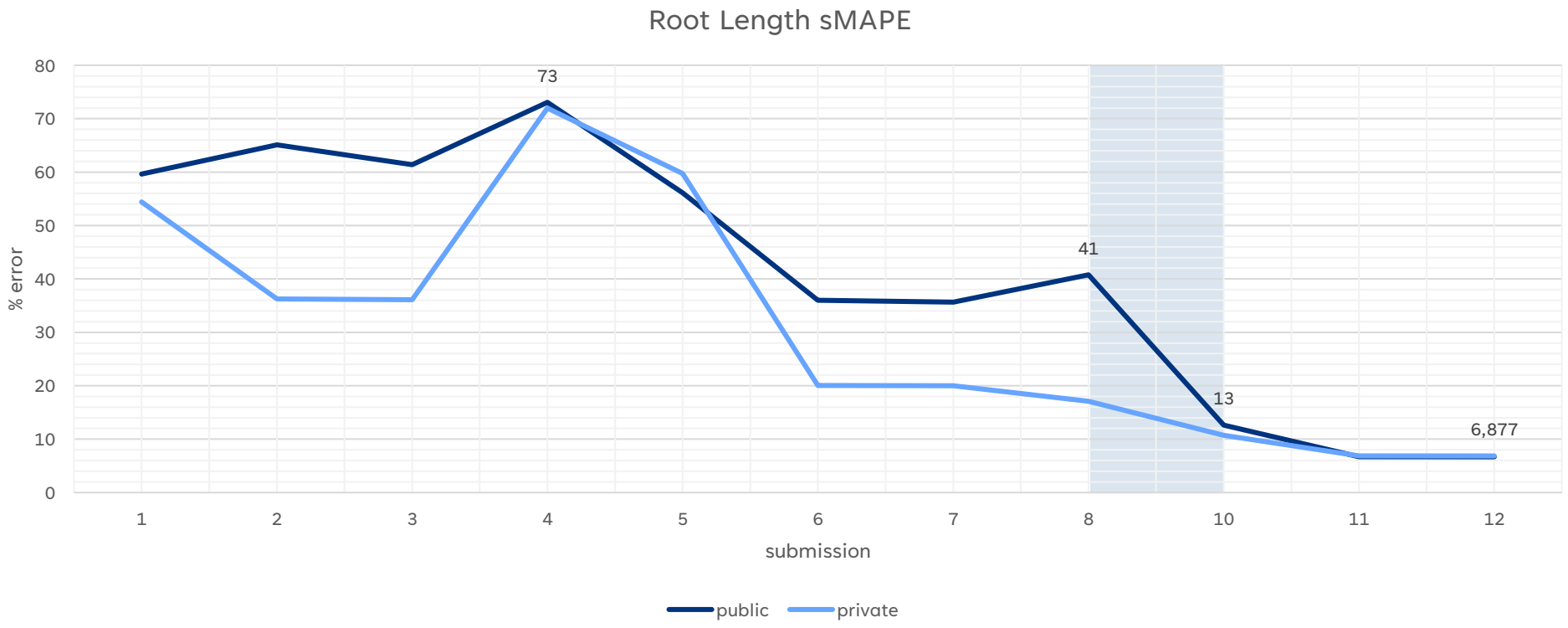
multi-class

Itr	Description	Loss	F1-score
2	Baseline	5.6364	0.2681
3	Added Square Root Weights	0.3444	0.6671
6	Filter out Background-Only Patches	0.0451	0.7628
24	Data Augmentation	0.0534	0.7705
39	Reduced Learning Rate on Plateau	0.0048	0.8308

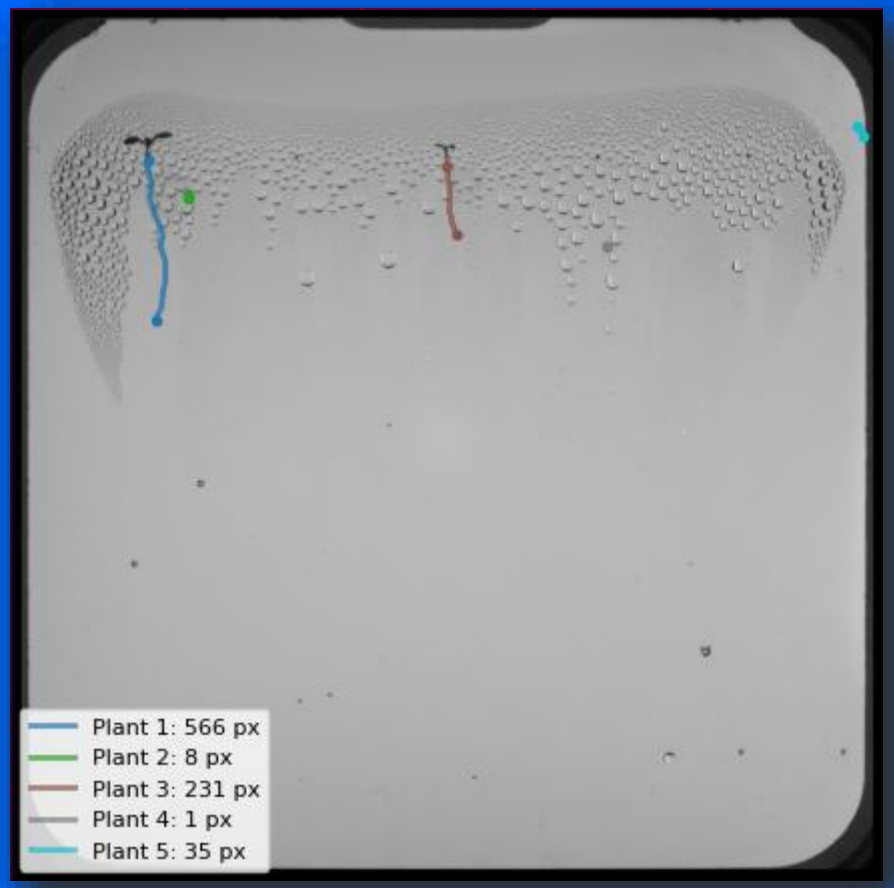
Kaggle Competition

12	Stijn Korver 242824		6.877

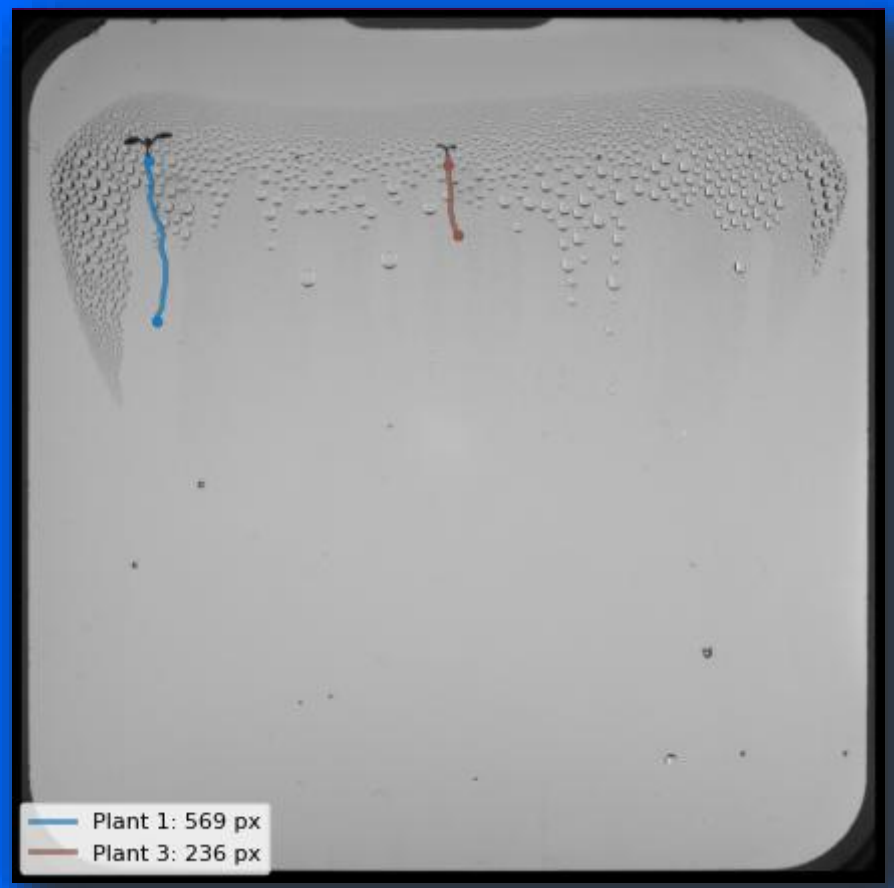
Kaggle Competition



Improving Instance Segmentation



before



after

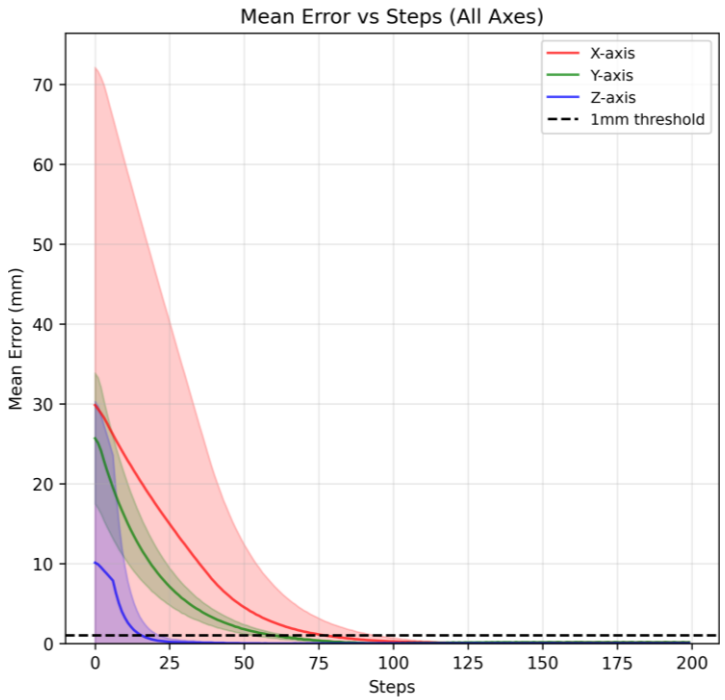
Robotics

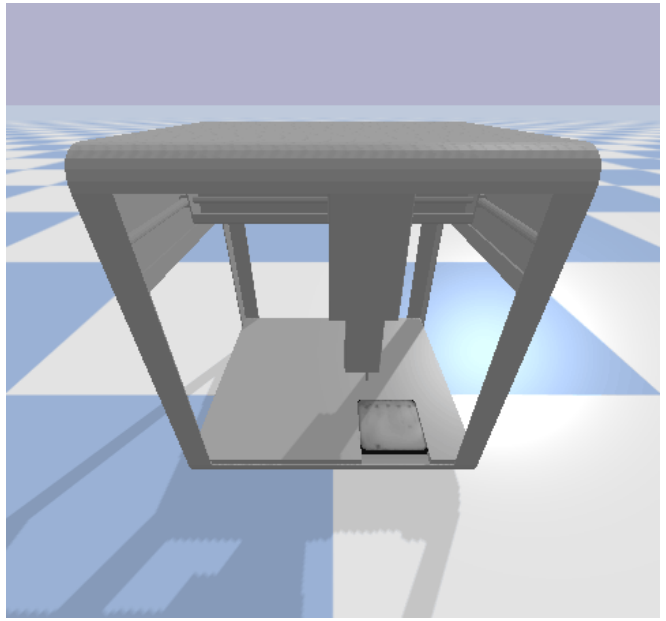
PID and RL

PID Controller

Proportional-Integral-Derivative

Parameter	Value
Proportional Gain (Kp)	15.0
Integral Gain (Ki)	0.5
Derivative Gain (Kd)	1.0

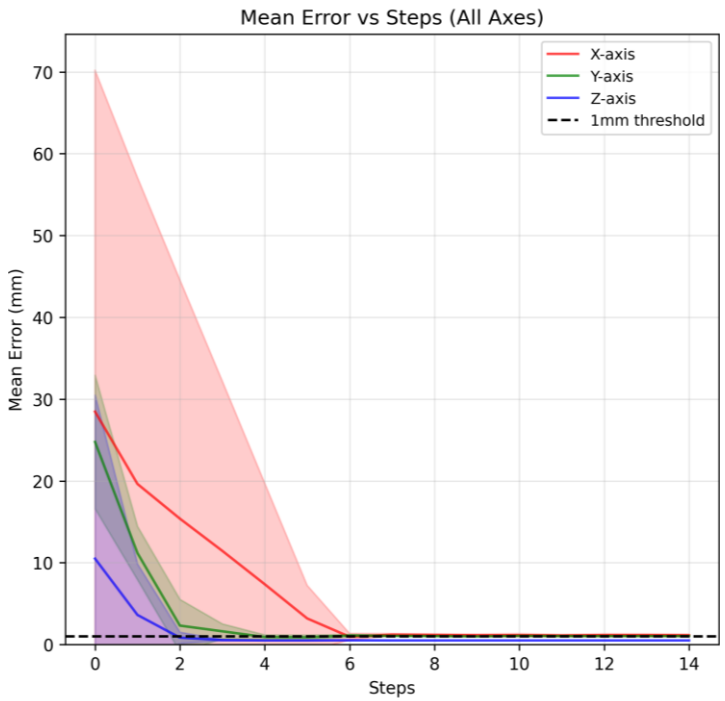


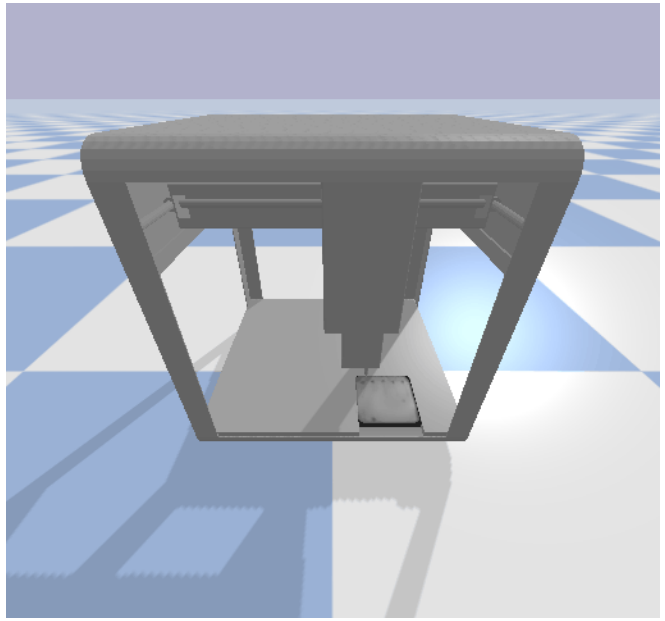


RL Controller

Reinforcement Learning

Parameter	Value
Distance Penalty	-distance
Boundary Penalty	-1.0
Success Bonus	+50.0





Benchmarking PID vs RL

Controller	Succes Rate	Settling Time	X-Error	Y-Error	Z-Error
PID	100%	70 steps	0.12 mm	0.15 mm	0.03 mm
RL	12%	3 steps	1.05 mm	0.98 mm	0.51 mm

average results of 50 targets per controller

Extracting Images from Simulation

```
npec.extract_tips

npec.run_controller
├── sim(num_agents=1)
├── sim.close()
```

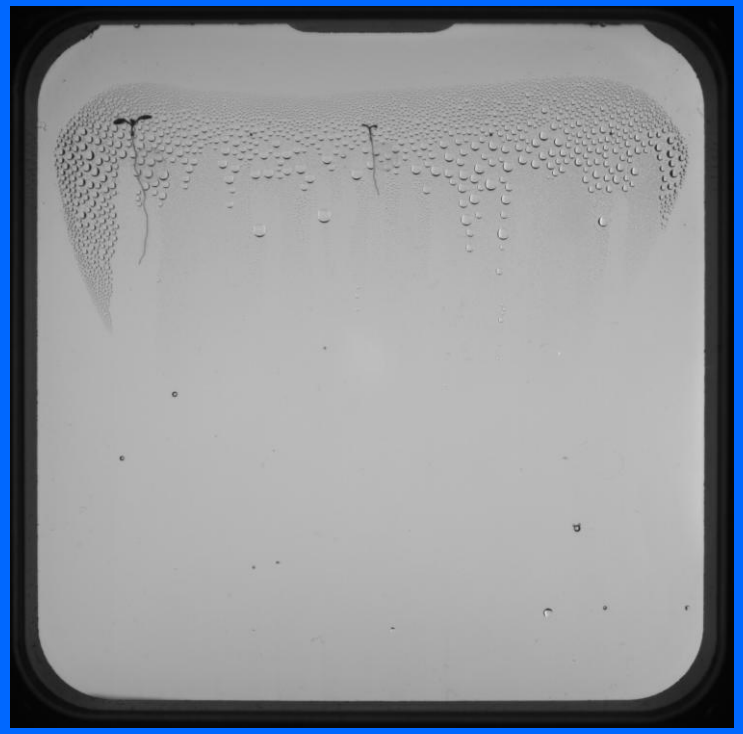
before

```
sim(num_agents=1)

npec.extract_tips

npec.run_controller

sim.close()
```

after

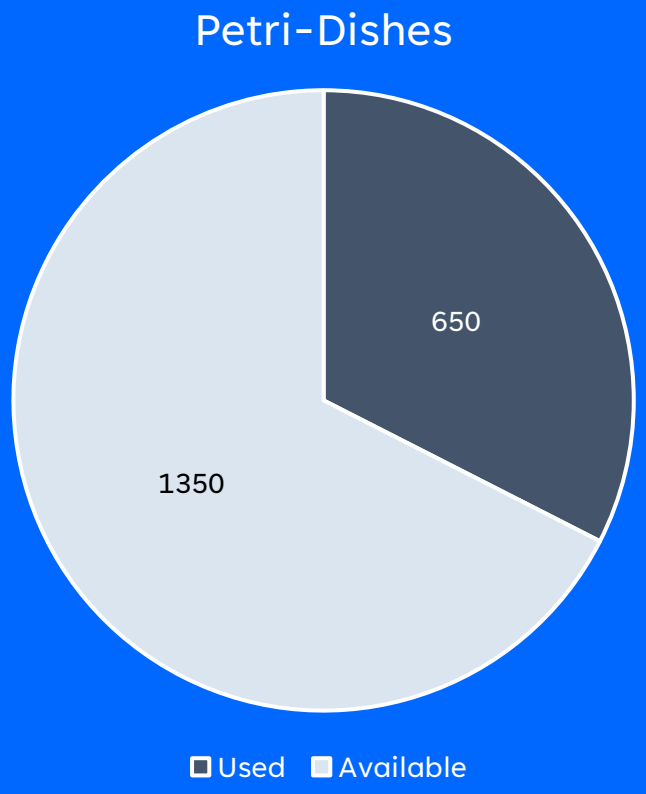
*not actual syntax

Assumptions



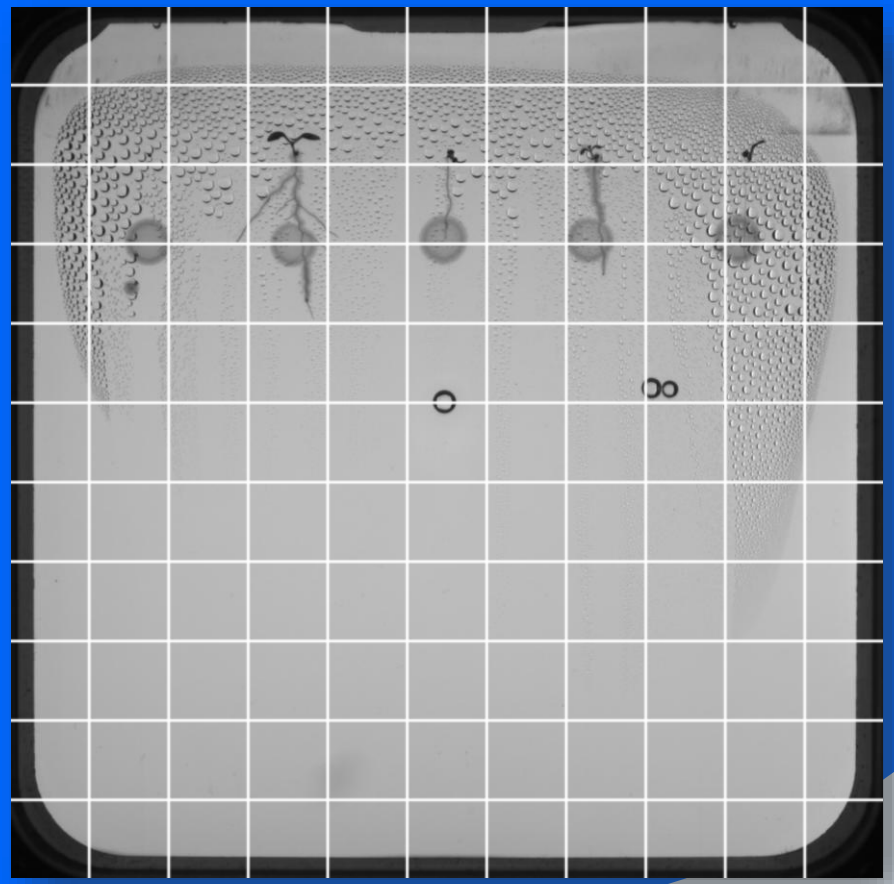
Limitations

- The Data that was Used
  - Amount
  - Variation
- Available Hardware
  - 60 hours of training
  - 39 iterations



Next Steps

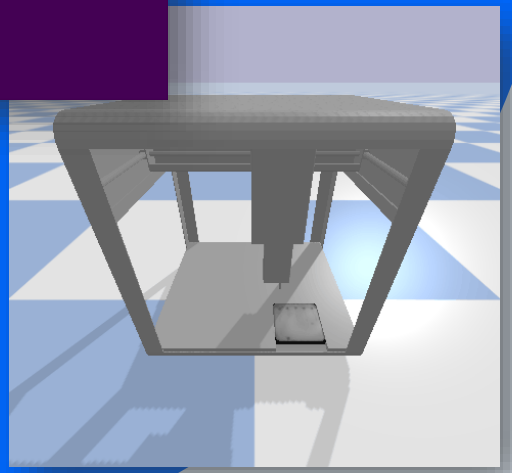
- Patch Size
- More Data
- Extensive Error Analysis
- Edge-Case Handling
- Seed Detection



256x256
patch size

Summary

- Reliable Root Length and Root Tip Detection
  - No Edge-Case Handling
  - Much Room for Improvement
- Reliable PID Controller
- Less Precise RL Controller



Thank you!

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